

Motor Condition During Movement of Joystick

Motor 4-Bit Data D3 D2 D1 D0				Motor Control Description (Robot Motion)
0	1	0	1	Forward
1	0	1	0	Reverse/backward
0	1	0	0	Turn - right - forward
0	0	0	1	Turn - left - forward
0	0	1	0	Turn - right - backward
1	0	0	0	Turn - left - backward
0	1	1	0	Rotate - right
1	0	0	1	Rotate - left
0	0	0	0	Stop

board without connecting RF modules. Ground pins of both the boards must be connected.

Now, power on both boards. Observe blinker LED2 on the transmitter board. It blinks whenever data bits are correctly transmitted in the loop. Blinker LED4 on the receiving board blinks whenever motor movement data is received.

Next, test movement of the motors using the joystick. If motors are not rotating, check voltage status of D3 to D0 (pins 2, 7, 6 and 5) of ATtiny13A at the receiving board using a voltmeter (or an LED with 2.2-kilo-ohm resistance in series).

Then, check connections of the motors and L293D. To change direction of rotation of the motors, interchange the wire connections on the receiving board as per requirement.

If everything is alright, power off both the boards. Remove the connecting wires between transmitter and receiver boards. Connect RF modules on the transmitter and receiver boards, and test for RF connectivity.

Fix the receiver unit on chassis of the robot along

PARTS LIST

Semiconductors:

- IC1, IC3 - LM1117-5.0 fixed 5V regulator
- IC2, IC4 - ATtiny13A MCU
- IC5 - L293D motor driver
- T1 - BC547 npn transistor
- D1, D2 - 1N4007 rectifier diode
- LED1-LED4 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R3, R4, R6, R8, R9 - 2.2-kilo-ohm
- R2, R5, R7 - 10-kilo-ohm

Capacitors:

- C1, C2, C6, C8 - 10 μ F, 25V electrolytic
- C3, C4, C7 - 0.1 μ F ceramic disk
- C5 - 100 μ F, 25V electrolytic

Miscellaneous:

- BATT.1 - 9V battery
- BATT.2 - 7.4V, 2200mAh lithium-ion battery
- CON1 - 5-pin connector
- CON2 - 3-pin connector
- M1, M2 - 5V - 7.4V battery-operated geared motor
- TX1 - 433MHz RF transmitter module
- RX1 - 433MHz RF receiver module
- S1 - Push-to-on switch
- S2 - On/off switch
- Joystick module
- Jumper wires
- 2-pin terminal connector (2 nos.)
- Robot cabinet assembly along with two rear wheels and one castor wheel

(All these are available from kitsnspares.com)

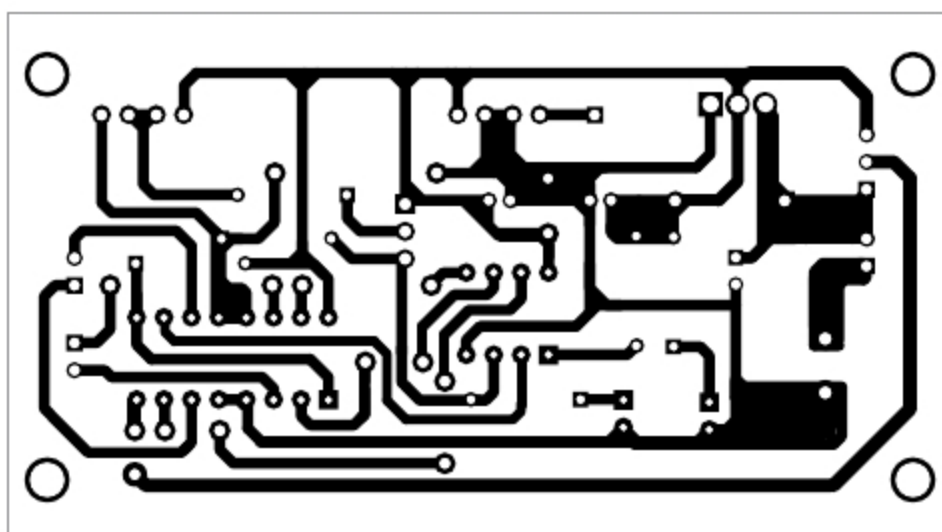


Fig. 8: Actual-size PCB layout of the receiver unit

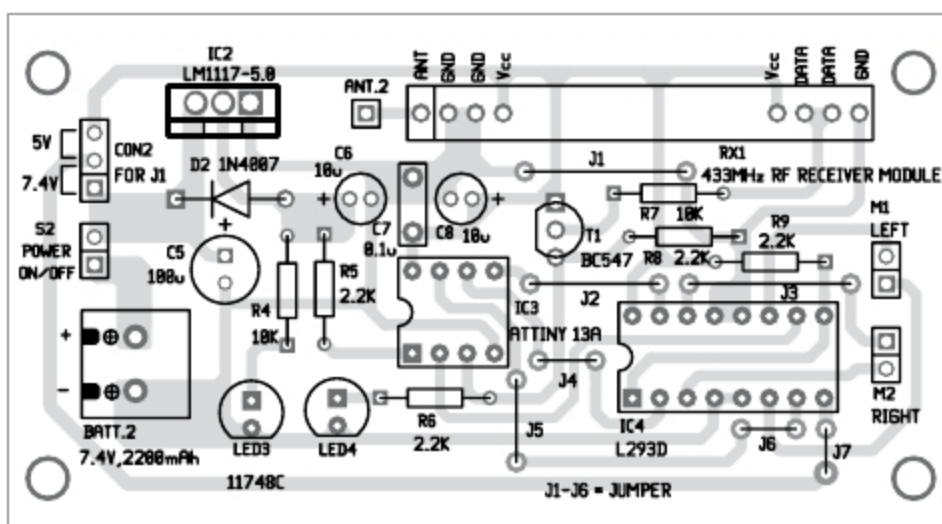


Fig. 9: Components layout for the receiver PCB

with the wheels. M1 and M2 are fixed on the chassis along with a castor wheel at the front of the robot. Keep the transmitter unit and the joystick with you to control the robot. Now, you can control the robot using the joystick from a distance of up to thirty metres; the range depends on RF modules. **EFY**



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